JEFFREY (JEB) SONG

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EDUCATION

University of Cambridge *2025-2026*
MASt in Mathematics

University of Washington (UW) *2023-2025*
B.Sc. in Physics (Honours), B.A. in Mathematics, 3.94/4.00.

National University of Singapore (NUS) *2021-2023*
B.Sc. in Physics, 4.36/5.00. *NUS College Programme (Honours College).*

PUBLICATIONS

J. Z. Song*, G. Kishony*, E. Berg, and M. S. Rudner, Vari-Cool: a non-unitary quantum variational protocol for simulated cooling, arXiv:2510.09749 (2025).

J. Z. Song, H. Ha, W. W. Ho, and V. B. Bulchandani, A theory of quasiballistic spin transport, arXiv:2503.15756 (2025).

RESEARCH EXPERIENCE

UW Prof. Mark Rudner Group — *Variational Cooling*
April 2025 - Present
Investigating variational cooling of the transverse field ising model (TFIM) on near-term quantum computers (IBM). We demonstrated cooling for 28 TFIM system qubits comparable to recent results by Google.

Rice University Asst. Prof. Vir Bulchandani Group — *Quasi-Ballistic Spin Transport*
Sep 2024 - Present
Analyzing regimes for long-lived ballistic modes of spin on anisotropic XXZ spin-$1/2$ chains. Contributed to analytical results and developed numerical simulations of these results. Our manuscript is currently under review as a letter in PRB.

NUS Asst. Prof. Ho Wenwei Group — *1D Measurement-Based Quantum Computation*
Jan 2023 - Present
Studied 1D Measurement-Based Quantum Computation (MBQC) using tensor networks for my UROPS project. Extended a recent 1D MBQC scheme (Stephen et al., 2022) to qutrits. My UROPS report was nominated for the best UROPS award by the Physics Department.

UW Assoc. Prof. Samu Taulu Group — *EEG Calibration via Quasi-Static Models*
June 2024 - June 2025
Developed and optimised EEG calibration algorithms informed by electromagnetic models, comparing both local and global approaches in `FieldTrip`. Simulated and analysed source localisation techniques. My research was reward the Mary Gate scholarship.

UW Prof. Arka Majumdar Group — *Optical Machine Learning*
Aug 2023 - Feb 2025
Explored multi-input reservoir computing (type of recurrent neural network) using optical spiral waveguides. Developed simulation scripts in Lumerical (`varFDTD` and `FDE`) to design geometries and process results.

*Equal contribution.

NUS Assoc. Prof. Kuldip Singh Group — *Foundations of Quantum Mechanics*
Jan 2023 - May 2023
Explored ψ-ontological models, Bell's inequality, and the Pusey-Barrett-Rudolph theorem for directed reading project. Completed a project paper and presentation on limitations of such models.

NUS Asst. Prof. Lee Chinghua Group — *Non-Hermitian Quantum Physics with Qiskit*
May 2022 - Feb 2023
Simulated quantum systems using IBM's `Qiskit` package. Reproduced the SSH model, implemented trotterization for arbitrary unitaries, and decomposed non-Hermitian Hamiltonians.

NUS Asst. Prof. Loh Huanqian Lab — *Lasers for Neutral Atom Quantum Computing*
Feb 2022 - June 2022
Built an intensity servo system and constructed optical setups involving AOMs and DDS for precise control. Presented an internal talk on error correction techniques for laser servos.

LEADERSHIP AND MENTORSHIP

UW Undergraduate Physics Mentoring Program *Jan 2024 - Present*
Mentor
Organized by the UW Society for Physics Students (SPS) Provided academic and personal support to three mentees, facilitating bi-weekly meetings and participating in academic and social events. I was on the panel for an information session for pre-majors on physics.

NUS College Philosophy Club *June 2022 - June 2023*
Founder
I organised bi-weekly philosophy discussions, with a focus on the philosophical aspects of science and math. I successfully organized several talks and lectures by philosophy professors, providing an opportunity for students to learn from experts in the field and expand their understanding of philosophy.

NUS Physics Society *August 2021 - August 2022*
Events Director
I helped organise a successful physics mentor-mentee program. Furthermore, I was instrumental in jointly planning and organizing online meetups for physics students, providing opportunities for networking and learning beyond the classroom.

AWARDS

Mary Gates Research Scholarship *2024*

Guy Pitzel Scholarship (Departmental Honours) *2024*

Goldwater Scholar UW Nominee *2024*

CRISP Award for Best UROPS Nominee *2023*

PRESENTATIONS

Improving EEG calibration using electrostatic multipole moments
UW Mary Gates Undergraduate Research Symposium *2025*

A Theory of Quasiballistic Spin Transport
UW SPS Research Symposium for Physical Sciences *2025*

TECHNICAL SKILLS

Technical Skills Python, Linux (Ubuntu/Fedora), Java, PyTorch, Lean, Lumerical, Qiskit, Matlab, Bash/Zsh Scripting, C